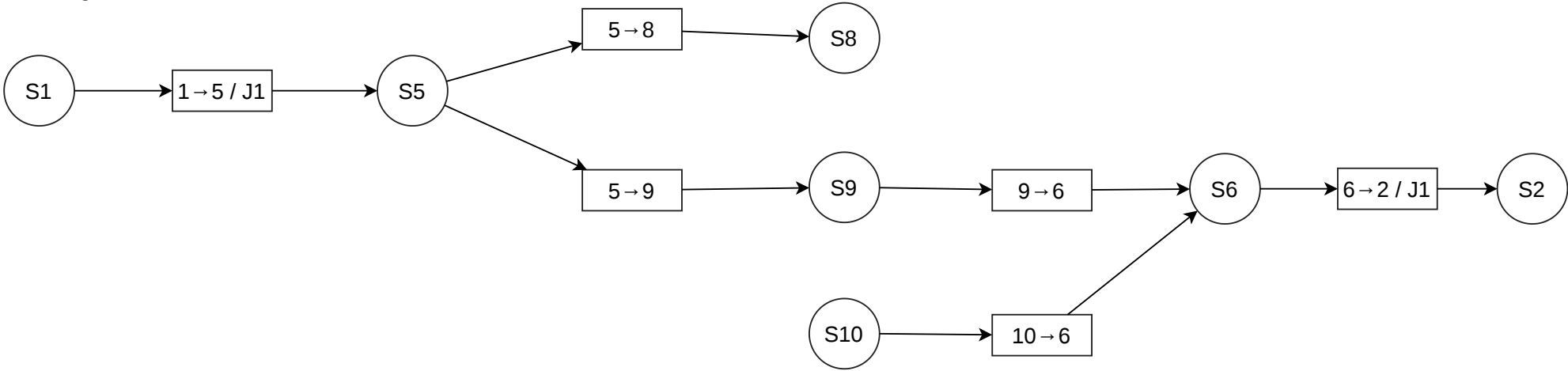


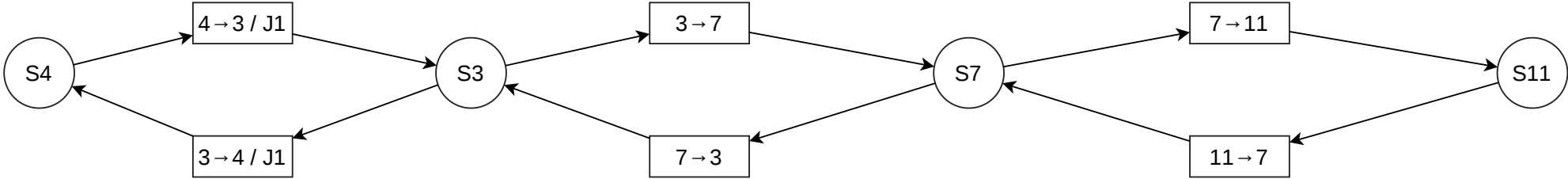
Railway Interlocking Petri Net v2 Final

Passenger



J2: 5→8, 5→9, 9→6, 10→6

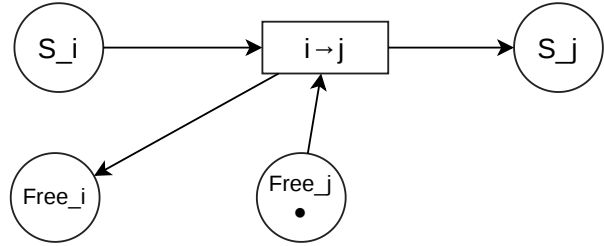
Freight



○ section (train token) □ move Route guard: follow assigned destination

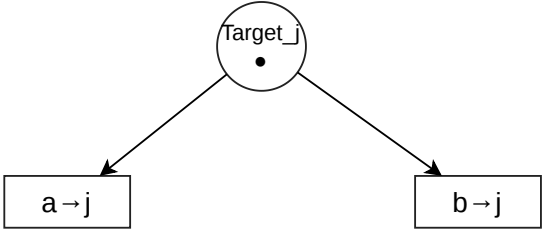
Control rules (guarded / coloured net)

Section capacity



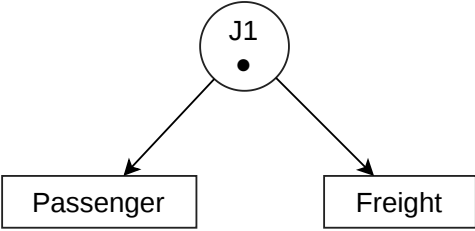
$M(S_i) \leq 1$; enter only if free.

Same target conflict



One target reservation per batch.

Passenger priority / J1



Freight guard: no eligible passenger J1 move.

J1 / J2: at most one move per batch. Reservations reset after batch.

Occupied target allowed only if its train leaves in the same batch; no cyclic swaps.

Free_j is returned when S_j is vacated. Priority is applied before final dependency pruning.

S9 shown once for both passenger routes. Tokens retain route / destination.